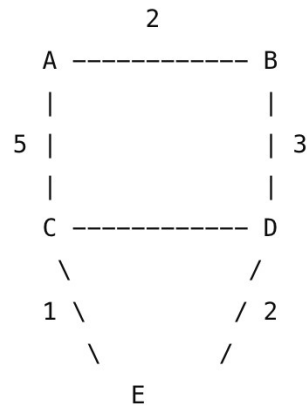


Exercise 2: Link State and Distance Vector Routing

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Learning objectives: Apply Dijkstra and Bellman-Ford; construct forwarding tables; analyze convergence, failures, and count-to-infinity.

Network topology:



Part 1 — Link State Routing

Question 1 — Dijkstra's Algorithm

Starting at router A, execute Dijkstra's algorithm. Show every update and the node confirmed at each step.

Step	Confirmed Node	D(B), Prev.	D(C), Prev.	D(D), Prev.	D(E), Prev.
0	A	2	5		
1	B	2	5	5	
2	C	2	5	5	6
3	D	2	5	5	6
4	E	2	5	5	6

Question 2 — Shortest-Path Tree

Draw the shortest-path tree rooted at A, then list the shortest path and total cost.

SPT sketch:

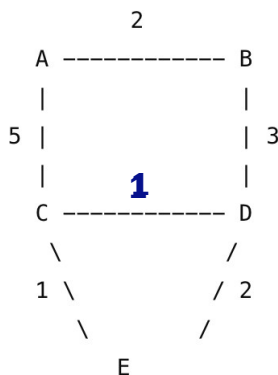
Destination	Shortest Path	Total Cost
B	A->B	A->B
C	A->C	A->C
D	A->B->D	A->B->D
E	A->C->E	A->C->E

Question 3 — Forwarding Table

Destination	Shortest Cost	Next Hop	Complete Path
B	2	B	A->B
C	5	C	A->C
D	5	B	A->B->D
E	6	C	A->C->E

Part 2 — Distance Vector Routing

Consider the same network. Initially, each router knows only the cost to its directly connected neighbors.



Question 4 — Bellman-Ford Update

Router A receives the following distance vectors from its neighbors:

From B:

A=2, B=0, C=4, D=3, E=5;

From C:

A=5, B=4, C=0, D=1, E=1.

Use the Bellman-Ford equation. Calculate router A's new distance to:

Show all calculations, including the neighbor through which the minimum-cost route is obtained.

Destination	Via B calculation	Via C calculation	Minimum Cost	Next Hop
B	2	9	2	B
C	6	5	5	C
D	5	6	5	B
E	7	6	6	C

Question 5 — Distance Vector Table

After receiving the distance vectors, construct router A's updated routing table.

Destination	Next Hop	Distance	Full Path
B	B	2	A->B
C	C	5	A->C
D	B	5	A->B->D
E	C	6	A->C->E

Part 3 — Routing Convergence

Question 6 — Link Failure

The C–D link fails. Explain separately what happens with Link State and Distance Vector routing.

Link State routing.

Your answer should describe:

- Which router(s) detect the failure.
- What information is flooded through the network.
- How routers update their link-state databases.
- How routers calculate new shortest paths.

Distance Vector routing.

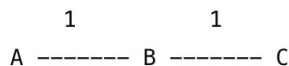
Your answer should describe:

- How routers learn about the failure.
- How updated distance vectors propagate.
- Why routers may temporarily have incorrect routing information.
- How routing eventually converges.

Link State	Distance Vector
C, D	Missing Hello or Keepalive messages.
flood LSA to another router to inform failure	Send updated distance vector to neighbors
check LSA, if new update databases	Update take time to propagate
Make new topology and run dijkstra's algorithm	Update until no router can improve its routing table

Part 4 — Count-to-Infinity

Consider



Initially A reaches C through B at cost 2. The B–C link fails, but B receives stale information from A saying C is reachable at cost 2.

Question 7

1. Why can this create a routing loop?

B find C is unreachable and keep ask A

A will keep inform B that C is reachable and so on

2. Show how the estimated distance to C can increase over successive iterations.

A-C = 2, B-C = 1. B-C fail A and B will keep reference cost from each other

3. What is count-to-infinity?

2 (first iteration)

2+1 = 3 Update A-C = 3

3+1 = 4 Update A-C = 4

and so on to inf

4. Why does it occur in Distance Vector routing?

DV Routing only knows destination, distance, next hop and depend on neighbors

5. Name a mitigation approach.

Split horizon

Part 5 — Compare LS and DV

Question 8

Characteristic	Link State	Distance Vector
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Basic algorithm	Build network map and SPF	learn distance from neighbors
Information maintained	Link-State DB	Routing Table / Distance Vector
Information exchanged	Link-State Advertisement	Distance Vecotr
Algorithm	SPF (Dijkstra's Algorithm)	Bellman-Ford Algorithm
Topology knowledge	know complete topology	only know neighbors
Routing updates	flood LSA through topology	send DV to neighbors hop by hop
Convergence	Fast	Slow
Routing loops	Less	More
Count-to-infinity	No	Vulnerable to count-to-inf
Example protocol	OSPF	RIP

Part 6 — Design Question

Question 9

You are designing a routing system for an enterprise with 500 routers requiring fast convergence, accurate topology knowledge, efficient routing, and hierarchical design. Choose LS or DV and justify your choice with at least three technical arguments. Your answer should discuss at least three advantages and/or disadvantages of your chosen approach.

LS

- 1.Fast convergence**
- 2.Less count-to-infinite problem**
- 3.If using OSPF. It will separate area which easier to scale**

Disadvantage

need more resources (more memory and more computation)
more complex

Challenge — Algorithmic Analysis

Question 10

For N routers and E links, compare LS and DV in information maintained, communication overhead, computation, convergence, and scalability. Explain why large-scale intra-domain routing generally favors Link State.

LS requires more memory to store per router $O(N+E)$ for linked state, DV is $O(N)$

LS may cost more overhead depends on topology complexity

DV computation is more simple by simply update DV with bellman-ford's equation while LS has to run SPF

LS has faster convergence (refer to Part 3, 4, 5)

Large scale domain requires more scalability which LS Protocol like OSPF can reduce flooding and SPF computation in bigger scale